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ASSET: A package for slit spectroscopy spectral extraction

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Software

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Summary

Based on the maximum a posteriori likelihood criterion, ASSET is a Julia package providing an optimized, robust spectral extraction method that can be used across various slit spectrographs, ensuring high-quality data extraction independent of the specific instrument. Thanks to its generic structures, different PSF and background models can be easily defined by the user to adapt the estimation process to the instrument. Fitting of the background, of the instrument's PSF and refinement of the spatio-spectral calibration of the detector are already possible if wanted.

Statement of need

Spectroscopy serves as a powerful tool for characterizing the chemical components of celestial bodies, including stars, planets and smaller objects in our solar system. Slit spectroscopy is a common approach particularly valuable for faint object observations that may otherwise be challenging with integral field spectroscopy. Many instruments have been developed in that regard. For example, the SPHERE-IRDIS instrument (Beuzit et al., 2019), with its near-infrared long-slit spectroscopy mode (Dohlen et al., 2008) covering J, H, and K bands (1 to 2.5 microns) with a resolution R=350, allows for detection and characterization of high contrast Dwarfs companions (Cheetham et al., 2018; Hinkley et al., 2015; Mesa et al., 2020) at small angular separation (0.5''). The James Web Space Telescope (JWST) contains two instruments equipped with slit mode : the NIRSpec (near-infrared) (Böker et al., 2023; Jakobsen et al., 2022), covering a wide range of 0.6 to 5.3 microns with low and high resolution, allowing for the characterization of faint solar system small bodies (Denneulin et al., 2023; Guilbert-Lepoutre et al., in prep.; Thomas et al., 2025) and faint stars or galaxies, and MIRI (mid-infrared) (Kendrew et al., 2015; Wright et al., 2023), covering a wavelength range of 5 to 14 microns, allows for the characterization of solar system objects (Müller et al., 2023) or, combined with a coronograph, the detection and characterization of exoplanets (Danielski et al., 2018; Henning et al., 2024). The future ELT instrument METIS should be equipped with a long-slit as well, allowing for the observation potential earth-like planet at very low separation (Maire et al., 2021).

To exploit the full potential of slit spectroscopy data, it is essential to develop advanced spectral extraction methods. Precise calibration of the spatio-spectral coordinates on the detector is crucial to map the spatial and spectral dimensions accurately. Additionally, accounting for the instrument's point spread function (PSF) is necessary to ensure reliable extraction of spectral feature. In many cases, robust background subtraction is also essential to eliminate unwanted contributions from other sources in the field of view, scattered light and detector artifacts such as bad-pixels or cosmic rays.

ASSET offers a flexible and thorough approach to address these challenges. This Julia package aims at providing a generic estimation framework based on the maximum a posteriori likelihood criterion. Different structures of parametric and non-parametric PSF are already proposed and can be refined via an alternate estimation scheme with the spectrum. With the same versatility, a custom background model can be defined, fitted and subtracted in the estimation scheme. Finally, the package includes several regularization structures via the use of `InverseProblem`.

The ASSET package is a full Julia implemented solution to extract spectrum from slit spectroscopy data. It yields an optimal extraction method (in the sense of the maximum of likelihood). It can be applied on any calibrated slit spectroscopy data regardless of the instrument used for the acquisition. It is designed to generalize the methods presented in (Thé et al., 2023) and (Denneulin et al., 2023) and to be fully adaptable to any slit spectroscopy instrument.

Estimation Framework

Slit spectroscopy acquisitions are two-dimensional data, consisting of one spatial and one spectral dimension. The most classical way to extract spectrum from spectroscopic data is the aperture extraction. In the context of slit data, it consists in integrating the photons, for each wavelength value in the spectral dimension, in a window of a given height. This method is suboptimal as it does not take into account the full width at half maximum (FWHM) of the instruments point spread function (PSF) or the noise statistic. Moreover, it can integrate artifacts (bad pixels, cosmic ray) that can pollute the spectral extraction.

Inverse problems framework is widely used in astrophysics to tackle these issues, such as Michalewicz et al. (2023) and Berdeu (2024).

The method used in the ASSET package was proposed by Thé et al. (2023) for the ESO/VLT SPHERE LSS, then adapted in Denneulin et al. (2023) for the JWST/NIRSpec FS. It requires the knowledge of the following maps as inputs:

- the L data maps $(d_\ell)_{\ell \in 1:L}$
- the L weights maps w_ℓ, which each element can be computed as the inverse variance of the pixel, forming the matrix $W_\ell = \Sigma_\ell^{-2}$. We assume that a defective pixel or artifacts have an infinite variance, i.e. a zero entry in W_ℓ.
- the L spatial distribution maps X_ℓ where 0 should correspond to the center of the studied object. #TODO: implement it, it should not be too long even if it seems a big deal
- the L spectral distribution maps Λ_ℓ.

Such maps must be stored in a `CalibratedData` structure using the constructor `CalibratedData(d, w, ρ_map, λ_map)`.

The outputs of the `extract_spectrum!` method are the extracted spectrum z, sampled over a given regular wavelength grid $(\lambda_n)_{n \in 1:N}$, and the parameters θ of the fitted PSF model, stored in a its corresponding structure. The spatial distribution maps X, stored in the `CalibratedData` structure, and the background map b, stored in a `BkgMdl` structure, are auto-calibrated in place.

ASSET uses an inverse problem approach to extract the spectra by fitting a chromatic PSF model on the data.

$$z, \theta \in \operatorname{argmin} \Big\{ \sum_\ell \|d_\ell - (m_\ell(z,\theta) + b)\|_{W_\ell}^2 + \mu_z \mathcal{R}_z(z) + \mu_\theta \mathcal{R}_\theta(\theta) + \mu_b \mathcal{R}_b(b), \Big\}$$

where $\mathcal{R}_z$, $\mathcal{R}_\theta$ and $\mathcal{R}_b$ are respectively the regularization of the extracted spectra, of the PSF parameters if required and of the background, with hyperparameters μ_z, μ_θ and μ_b.

$$m_\ell(z,\theta) = \alpha_\ell Z(\Lambda_\ell, z) \odot H(\theta, X_\ell, \Lambda_\ell)$$

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is the model of the data. It is the Hadamard (element-wise) product of the spectrum interpolated in the camera plane

$$Z(\Lambda, z)_{j,k} = \sum_n \phi\Big(\frac{\Lambda_{j,k} - \lambda_n}{\delta_\lambda}\Big) z_n$$

with ϕ an interpolation kernel, and of the chromatic PSF H. This PSF can have different degrees of freedom depending of the chosen model. In `ASSET` framework, we distinguish `ParametricPSF` , with low degree of freedom, and `NonParametricPSF` , with hight degree of freedom. A `ParametricPSF` H is a function parametrized by a few unknown variables θ_n, *e.g.* a Gaussian chromatic with a minimum width model:

$$H(\theta, X_\ell, \Lambda_\ell)_{j,k} = \big(2\pi(\theta_1 \Lambda_{j,k\ell}^2 + \theta_2)\big)^{-1} \exp\Big(-\frac{X_{j,k,\ell}^2}{2(\theta_1 \Lambda_{j,k,\ell}^2 + \theta_2)}\Big).$$

It does not require any regularization. See Thé et al. (2023) and Denneulin et al. (2023) for more details.

A `NonParametricPSF` is parametrized directly by some profile $(\theta_m)_{m\in 1:M}$, *e.g.* taking o order of the speckles expansion model (Devaney & Thiébaut, 2017) which is the interpolation of the profiles θ_o in a reference plane of the spatial coordinates $(x_m)_{m\in 1:M}$:

$$H(\theta, X_\ell, \Lambda_\ell)_{j,k} = \sum_o \gamma(\Lambda_\ell)^o \sum_m \phi\Big(\frac{\gamma(\Lambda_\ell) X_\ell - x_m}{\delta x}\Big) \theta_{m,o}(x_m).$$

For such a PSF, θ must be regularized. The regularization type and the initial hyperparameter must be given in the `NonParametricPSF` structure. The hyperparameter is auto-calibrated in the method, see Thé et al. (2023) for and references therein for more details.

The package provide several `ParametricPSF` and `NonParametricPSF` and the users can easily implement their own.

Usage Examples

In these example, ASSET is applied to the target GSPC P 330 E (Program ID 1538) observed with the JWST NIRSpec and MIRI instruments. Being a G dwarfs reference star, the reference spectrum is available (Bohlin et al., 2014) and is used to demonstrate the capacities of the package to provide a good extraction. For each example, the reference spectrum is resampled to the resolution of the extracted spectra.

JWST/NIRSpec Fixed Slit

These data were observed the 08/30/2022 with the S1600A1 Fixed Slit with the PRISM grating and CLEAR filter. It is composed of 5 dithers.

NIRSpec Fixed Slit data have the particularity that under 3.25μm the Full Width at Half Maximum (FWHM) of the PSF is smaller than the pixel width, thus chromatic PSF is "blured" by the pixel. Thus a chromatic PSF with a minimum width (corresponding to the `chromwmwPSF` particule) should be use. The Figure 1 and Figure 2 present the extracted spectrum and fitted PSF models for a gaussian PSF `chromwmwGaussianPSF`, a Moffat PSF `chromwmwMoffatPSF` and one order of the speckle expansion PSF `SeriesExpansionPSF`. The profile fitted by the `SeriesExpansionPSF` is more precise, however the slope of the spectra is affected by the fact that these PSF model does not account for the PSF blur, which is an ongoing work. The parametric PSF account for this blur and allow for a good extraction of the spectrum. ASSET spectral extraction is also more robust to outliers compared to the JWST pipeline extactions.

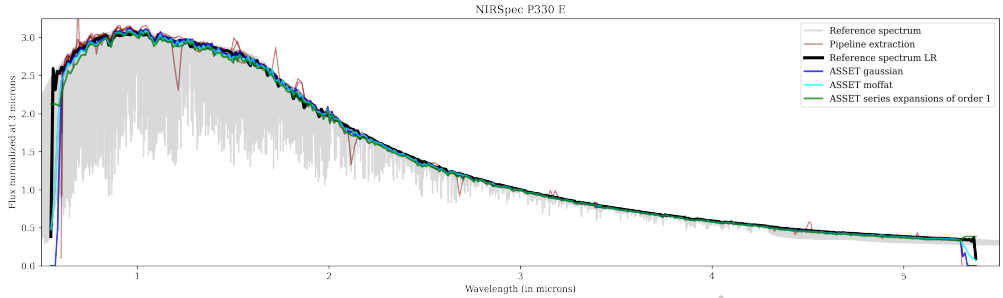

Figure 1: Comparison of the spectra extracted with the pipeline and ASSET for different PSF models

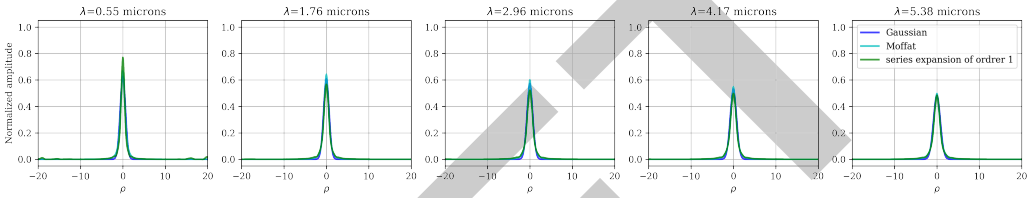

Figure 2: Comparison of the shape of the auto-calibrated PSF for each ASSET extracted spectrea

JWST/MIRI Long Slit

These data were observed the 08/14/2022 with the MIRI LRS Slit and the P750L filter. It is composed of 2 dithers. For the bandwidth covered with the MIRI LRS Slit, the target is very faint hence a very bright background for the largest wavelength is present due to a longer integration time.

The Figure 3 and Figure 4 present the extracted spectrum and fitted PSF models for a gaussian PSF `chromGaussianPSF`, a Moffat PSF `chromMoffatPSF`, and one and two orders of the speckle expansion PSF `SeriesExpansionPSF`. The profile fitted by the `SeriesExpansionPSF` is more precise, and the slop fits more accurately the reference spectrum around 5 μm. ASSET is more robust to the background brightness than the pipeline method, but it remains perfectible.

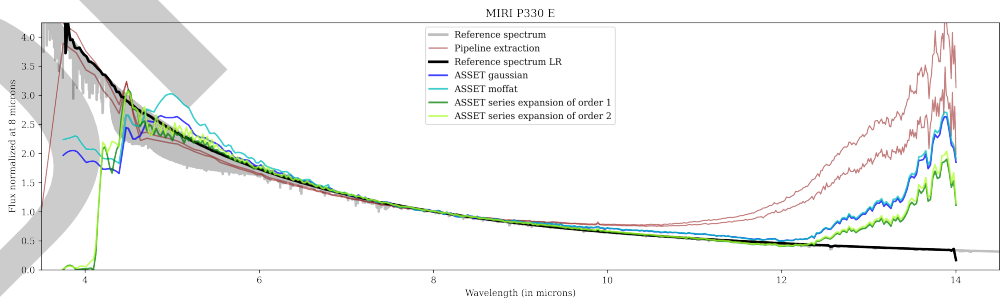

Figure 3: Comparison of the spectra extracted with the pipeline and ASSET for different PSF models

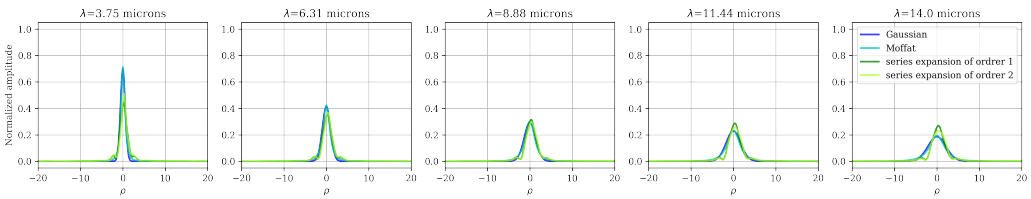

Figure 4: Comparison of the shape of the auto-calibrated PSF for each ASSET extracted spectra

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Research projects using the package

The ASSET package can be used for many slit spectrograph, such as the SPHERE/IRDIS-LSS (Thé et al., 2023) used to characterize exoplanets. In this context, speckles are forming a high-contrasted structural background where the extraction of the planet's spectrum is achieved by the method's joint estimation of this background and the instrument's PSF.

The package is also currently used to extract spectra from JWST/NIRSpec and MIRI data (Denneulin et al., 2023; Guilbert-Lepoutre et al., in prep.). These two instruments involves a diverse set of slits, spectral resolution and positions on the detector, to observe a vast range of targets in terms of flux. The flexible and multi-frame approach of ASSET is particularly interesting as it provides a single methodology to all these problems.

A particular interest in ongoing work is to correctly extract the spectrum of interest from the strong, but smooth, background present in some MIRI data. NIRSpec data also present some challenges at lower wavelength, where the instrumental PSF is blurred by the pixel integration. Finally, we aim to generalized such an approach to Integral Field Units data.

Dependencies

`PointSpreadFunctions`, `InverseProblems`, `InterpolationKernels`, `LazyAlgebra`, `LinearInterpolators`, `OptimPackNextGen`, `PowellMethods`, `AMORS` and `SparseArrays`.

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